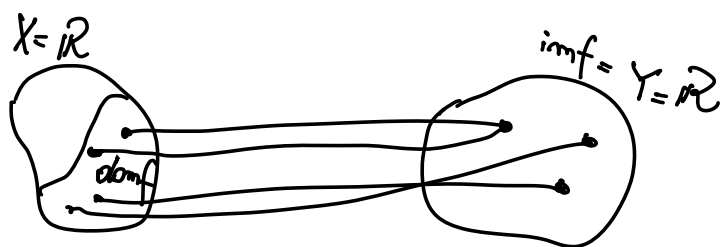


Def $f: \text{dom} f \subseteq X = \mathbb{R} \rightarrow Y = \mathbb{R}$

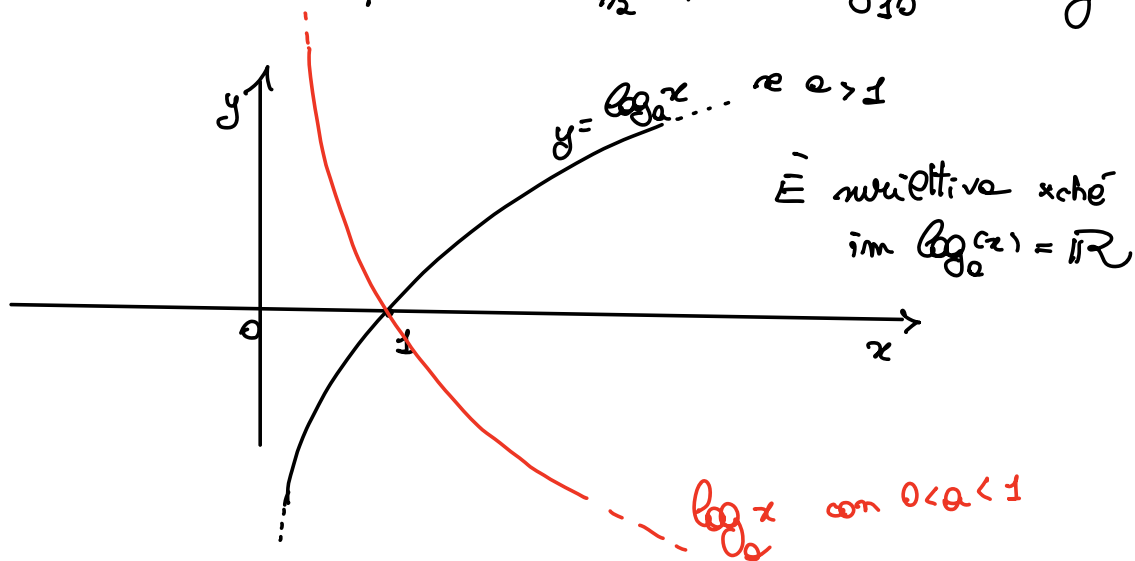
f suriettiva $\Leftrightarrow \text{im} f = Y (= \mathbb{R})$



cioè $\forall y \in Y \quad \exists x \in \text{dom} f : y = f(x)$

ES) $f(x) = \log x = \log_e x = \ln x$

altrimenti: $\log_7(x)$, $\log_{1/2}(x)$, $\log_{10}(x) = \text{Log}(x)$



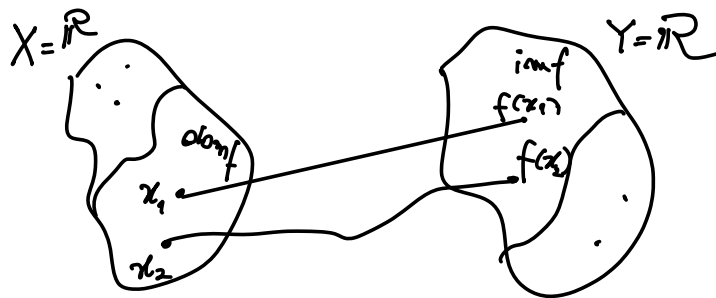
Def $f: \text{dom} f \subseteq X = \mathbb{R} \rightarrow Y = \mathbb{R}$

f iniettiva $\Leftrightarrow \forall x_1, x_2 \in \text{dom} f$

$$x_1 \neq x_2 \Rightarrow f(x_1) \neq f(x_2)$$

$$(P \Rightarrow Q \Leftrightarrow \neg Q \Rightarrow \neg P)$$

$$f(x_1) = f(x_2) \Rightarrow x_1 = x_2$$



$$\text{cioè } \forall y \in \text{imf} \quad \underbrace{\exists!}_{\substack{\text{esiste} \\ \text{ed è} \\ \text{unico}}} x \in \text{domf} : y = f(x)$$

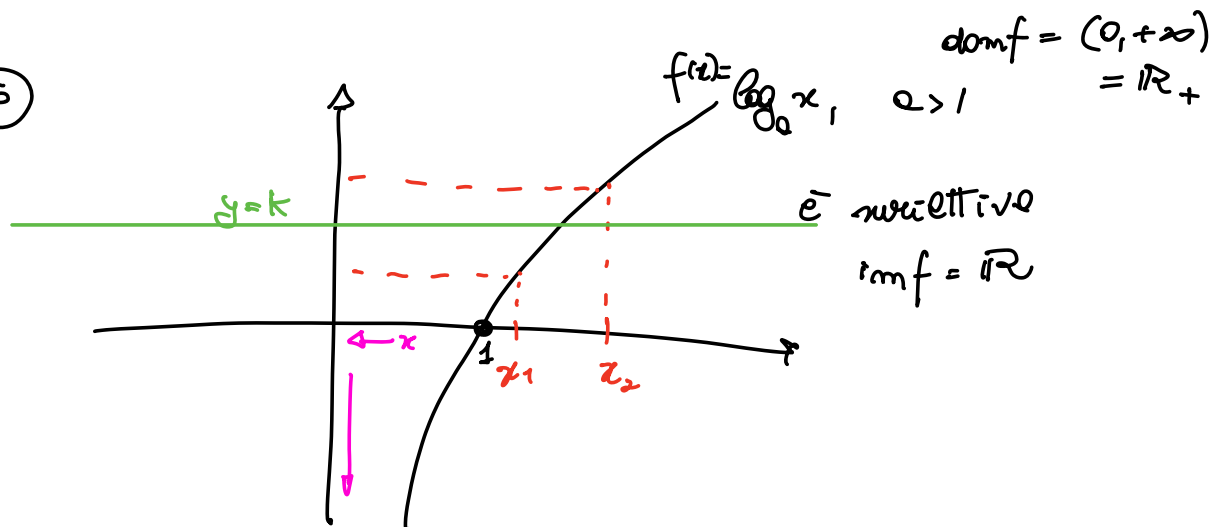
Def f iniettiva su $A \subseteq \text{domf}$ se
 $f|_A : A \rightarrow \mathbb{R}$ è iniettiva

Def $f : \text{domf} \subseteq X = \mathbb{R} \rightarrow Y = \mathbb{R}$

f biettiva se f è iniettiva e suriettiva

(ES) $f(x) = mx + q$, $m \neq 0$ $f : \mathbb{R} \rightarrow \mathbb{R}$
 è biettiva $\iff$ $\begin{cases} \text{è iniettiva} \\ \text{imf} = \mathbb{R} \end{cases}$

ES



$\forall k \in \mathbb{R}$ la retta $y=k$ interseca $\Gamma(f)$

$\Rightarrow f$ è suriettiva

$$\Gamma(f) \cap \Gamma(\text{retta}) \neq \emptyset$$

ES

$$f(x) = x^m$$

m dispari ($m=1, 3, 5, \dots$)

$$\text{dom } f = \mathbb{R}$$

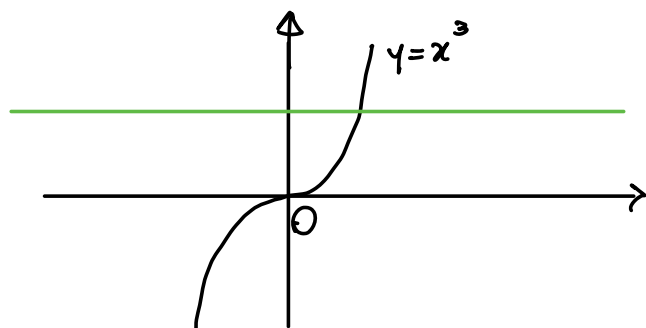
$$\text{im } f = \mathbb{R}$$

$\Downarrow$
 f è suriettiva

f è anche iniettiva

$$\Rightarrow f: \mathbb{R} \rightarrow \mathbb{R}$$

biettiva



f è dispari

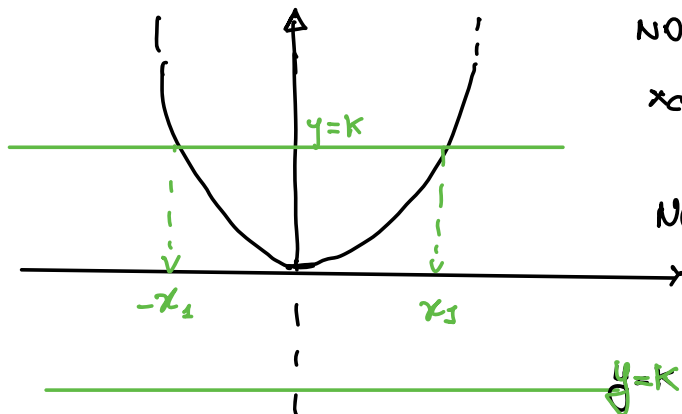
$$f(-x) = -f(x)$$

simmetrica rispetto O

n pari ($n=2, 4, 6, \dots$)

$n=2$

$$y=x^2$$



NON è suriettiva

$$\text{range } f = [0, +\infty) \neq \mathbb{R}$$

NON è iniettiva

↓
perché è pari

gli intervalli I massimali t.c. $f|_I$ è iniettiva
sono $(-\infty, 0]$ e $[0, +\infty)$

ES $f(x) = \log(2-|x|)$

$$\text{dom } f = \{x \in \mathbb{R} : 2-|x| > 0\} = (-2, 2)$$

$$|x| < 2 \Leftrightarrow -2 < x < 2$$

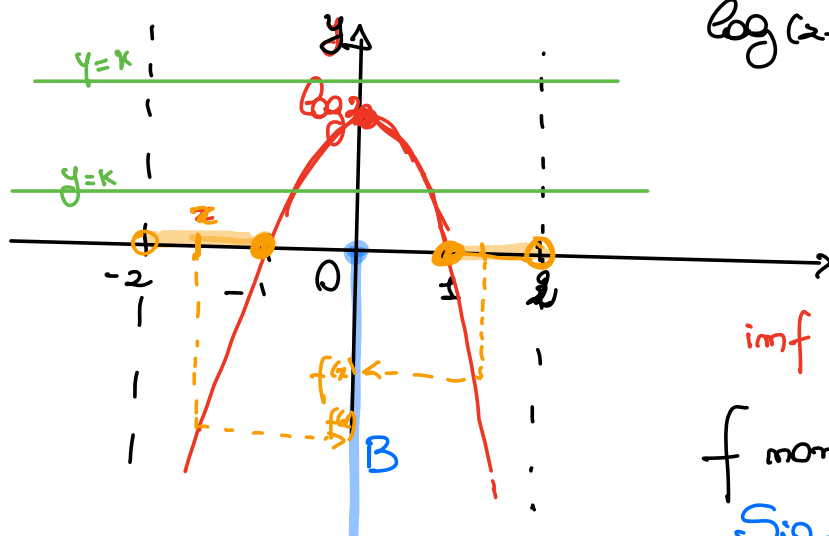
f è pari perché $\forall x \in \text{dom } f, f(-x) = f(x)$

$$f(-x) = \log(2 - \underbrace{|-x|}_{|x|}) = \log(2 - |x|) = f(x)$$

$\Rightarrow f$ NON è iniettiva

essendo pari mi basta studiare per $x \geq 0$ (poi
simmetria rispetto all'asse y)

$x \geq 0$: $f(x) = \log(2-x)$ $0 \leq x < 2$ $f(0) = \log 2$ y



$$\log(2-x) = 0 \Leftrightarrow 2-x = 1 \Leftrightarrow x = 1$$

$\text{imf} = (-\infty, \log 2]$

f non è suriettiva

Sia $B = (-\infty, 0] \subseteq \text{imf}$

$\Rightarrow f^{-1}(B) \subseteq \text{domf}$

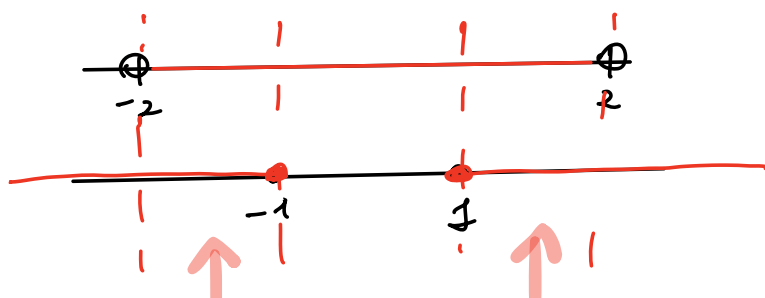
$f^{-1}(\underbrace{(-\infty, 0]}_B) \stackrel{x \text{ definizione}}{=} \{x \in \text{domf} : f(x) \in \underbrace{(-\infty, 0]}_B\}$

$= \{x \in (-2, 2) : f(x) \leq 0\}$

$= \{x \in (-2, 2) : \log(2 - |x|) \leq 0\}$

$$\begin{cases} -2 < x < 2 \\ \log(2 - |x|) \leq 0 \end{cases} \quad \begin{cases} -2 < x < 2 \\ 2 - |x| \leq 1 \end{cases} \quad \begin{cases} -2 < x < 2 \\ |x| \geq 1 \end{cases}$$

$$\begin{cases} -2 < x < 2 \\ x \geq 1 \vee x \leq -1 \end{cases}$$



$$f^{-1}((-\infty, 0]) = (-2, -1] \cup [1, 2)$$

Operazioni su funzioni

$$f: \text{dom } f \subseteq \mathbb{R} \rightarrow \mathbb{R}$$

$$g: \text{dom } g \subseteq \mathbb{R} \rightarrow \mathbb{R}$$

$$\alpha, \beta \in \mathbb{R}$$

$$\bullet \underbrace{(\alpha f + \beta g)(x)}_{\text{combinazione lineare di } f \text{ e } g} \stackrel{\text{def}}{=} \alpha f(x) + \beta g(x)$$

$$\forall x \in \text{dom } f \cap \text{dom } g$$

$$f(x) + g(x) = \sin x + \sqrt{x}$$

$$\begin{aligned} \forall x \in \text{dom } f \cap \text{dom } g &= \\ &= \mathbb{R} \cap [0, +\infty) \\ &= [0, +\infty) \end{aligned}$$

$$\bullet (fg)(x) \stackrel{\text{def}}{:=} f(x) g(x)$$

$$\forall x \in \text{dom } f \cap \text{dom } g$$

$$\bullet \left(\frac{f}{g}\right)(x) \stackrel{\text{def}}{=} \frac{f(x)}{g(x)}$$

$$\forall x \in \text{dom } f \cap \{x \in \text{dom } g: g(x) \neq 0\}$$

N.B. $f(x) = 1 \quad \forall x \in \mathbb{R}$ $\overset{\text{funzione}}{\downarrow}$ $f =$ costante pari a 1

$$\left(\frac{1}{g}\right)(x) = \frac{1}{g(x)}$$

è la reciproca di g

$$\text{e.t.c.} \quad g(x) \cdot \frac{1}{g(x)} = 1$$

es) $f(x) = x^2 + 1, \quad \text{dom } f = \mathbb{R}$

$$\left(\frac{1}{f}\right)(x) = \frac{1}{x^2 + 1} \quad \text{dom } f = \mathbb{R}$$

↑ è la reciproca di f

MA VEDREMO che f non è invertibile

(non ammette funzione inversa)

Composizione di funzioni

$$f: \text{dom } f \subseteq \mathbb{R} \rightarrow \mathbb{R}, \quad g: \text{dom } g \subseteq \mathbb{R} \rightarrow \mathbb{R}$$

$$(g \circ f)(x) := g(f(x))$$

$$\begin{array}{ccc} x & \xrightarrow{f} & f(x) \\ \uparrow \text{dom } f & & \uparrow \\ y & \xrightarrow{g} & g(y) = g(f(x)) \end{array}$$

$$g \circ f : \text{dom}(g \circ f) \subseteq \mathbb{R} \rightarrow \mathbb{R}$$

$$\underline{\text{dom}(g \circ f) = \{x \in \text{dom} f : f(x) \in \text{dom} g\} \subseteq \text{dom} f}$$

$$x \in \text{dom}(g \circ f) \Leftrightarrow \begin{cases} x \in \text{dom} f \\ f(x) \in \text{dom} g \end{cases}$$

$\text{im}(g \circ f) \subseteq \text{im} g$ perché g "trasporta" solo gli $y \in \text{dom} g$ che sono del tipo $y = f(x)$

N.B. $g \circ f \neq f \circ g$

(ES) $f(x) = x^2 - 1$, $g(x) = \frac{1}{|x+1|}$

$\text{dom} f = \mathbb{R}$, $\text{dom} g = \mathbb{R} \setminus \{-1\}$, $\text{im} g = \mathbb{R}_+$
 $\text{im} f = [-1, +\infty)$

$$\begin{aligned} \text{dom}(g \circ f) &\stackrel{\text{def}}{=} \{x \in \text{dom} f : f(x) \in \text{dom} g\} \\ &= \{x \in \mathbb{R} : x^2 - 1 \neq -1\} \\ &= \{x \in \mathbb{R} : x^2 \neq 0\} = \mathbb{R} \setminus \{0\} \end{aligned}$$

$$(g \circ f)(x) = g(f(x)) = g(\underbrace{x^2 - 1}_t) = \frac{1}{|x^2 - 1 + 1|} = \frac{1}{|x^2|} = \frac{1}{x^2}$$

$t = x^2 - 1$ $g(t) = \frac{1}{|t+1|}$

$\text{im}(g \circ f) = \mathbb{R}_+ = (0, +\infty)$

$$\begin{aligned}\text{dom}(f \circ g) &= \{x \in \text{dom } g : g(x) \in \text{dom } f\} \\ &= \{x \in \mathbb{R}, x \neq -1 : \underbrace{\frac{1}{|x+1|}}_{\text{OK}} \in \mathbb{R}\} = \mathbb{R} \setminus \{-1\}\end{aligned}$$

$$\begin{aligned}(f \circ g)(x) &= f(g(x)) = f\left(\frac{1}{|x+1|}\right) = \left(\frac{1}{|x+1|}\right)^2 - 1 \\ &= \frac{1}{(x+1)^2} - 1\end{aligned}$$

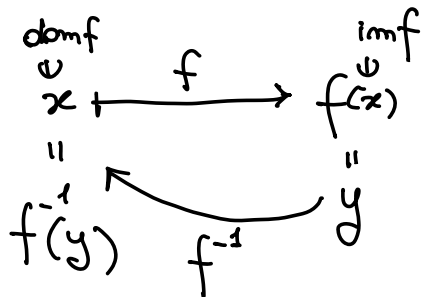
$t = \frac{1}{|x+1|}$
 $f(t) = t^2 - 1$

$$\text{im}(f \circ g) = (-1, +\infty) \neq \text{im } f = [-1, +\infty)$$

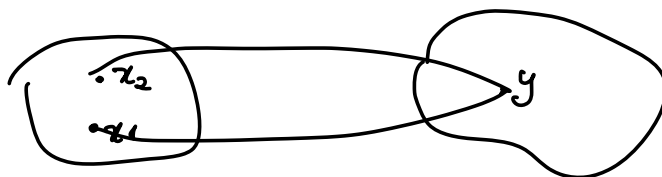
Funzione inversa

Sia f una funzione iniettiva

$$f: \text{dom } f \subseteq \mathbb{R} \rightarrow \mathbb{R}$$



$$\begin{aligned}\text{dom } f^{-1} &= \text{im } f \\ \text{im } f^{-1} &= \text{dom } f\end{aligned}$$



l'inversa f^{-1}
 $\nexists$ perché f non
 è iniettiva

Sia $f: \text{dom} f \subseteq \mathbb{R} \rightarrow \mathbb{R}$ iniettiva

L'inversa di f , denotata con f^{-1} , è una fz

$$f^{-1}: \text{dom} f^{-1} \subseteq \mathbb{R} \rightarrow \mathbb{R} \text{ t.c.}$$

$$1) \quad \text{dom} f^{-1} = \text{im} f$$

$$2) \quad \text{im} f^{-1} = \text{dom} f$$

$$3) \quad f^{-1}(f(x)) = x \quad \forall x \in \text{dom} f$$

$$4) \quad f(f^{-1}(y)) = y \quad \forall y \in \text{im} f$$

N.B. f^{-1} associa ad ogni $y \in \text{im} f$

l'unico $x \in \text{dom} f$ t.c. $f(x) = y$

$$f^{-1}(y) = x \iff y = f(x) \quad \forall x \in \text{dom} f \\ \forall y \in \text{im} f$$

N.B. f iniettiva $\Leftrightarrow f^{-1}$ invertibile

N.B. Se f è invertibile $\Rightarrow f^{-1}$ è
invertibile e $(f^{-1})^{-1} = f$

(ES) $f(x) = x^2$ non è invertibile
(su $\text{dom} f$)

perché non è iniettiva (è PARI!)

Ma $f|_{(-\infty, 0]}$ e $f|_{[0, +\infty)}$ sono

invertibili perché iniettive

Come si passa da $M(f)$ a $M(f^{-1})$?

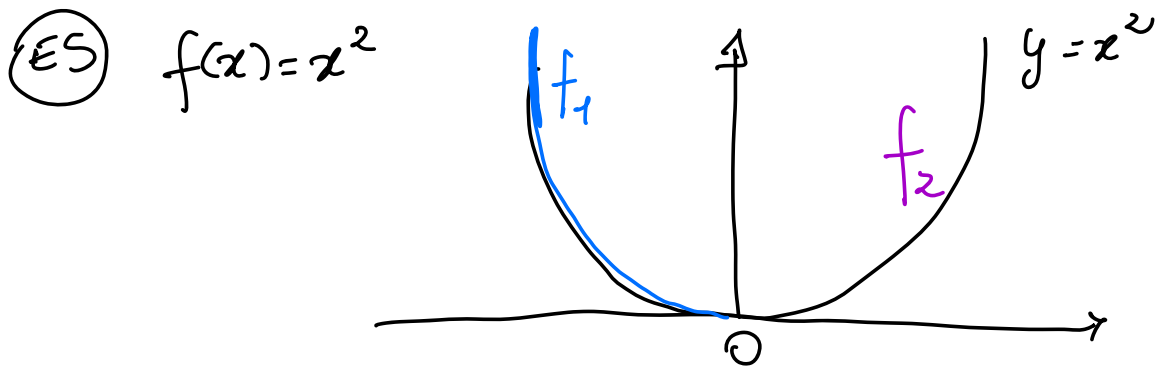
Una la simmetria rispetto $y = x$

Perché:

$$\begin{aligned} M(f) &= \{ (x, \underbrace{f(x)}_y) \mid x \in \text{dom} f \} \\ &= \{ (f^{-1}(y), y) \mid y \in \text{im} f \} \end{aligned}$$

$$f^{-1} = \{ (y, \underline{f^{-1}(y)}) , y \in \text{dom } f^{-1} = \text{im } f \}$$

invertire le coordinate significa fare una simmetria rispetto $y=x$



$$f_1 = f|_{(-\infty, 0]} : (-\infty, 0] \xrightarrow{f_1} \text{im } f_1 = [0, +\infty)$$

$$\begin{array}{ccc} x & \xrightarrow{\quad} & f_1(x) = x^2 \\ \parallel & & \parallel \\ -\sqrt{y} & \xleftarrow{f_1^{-1}} & y \end{array}$$

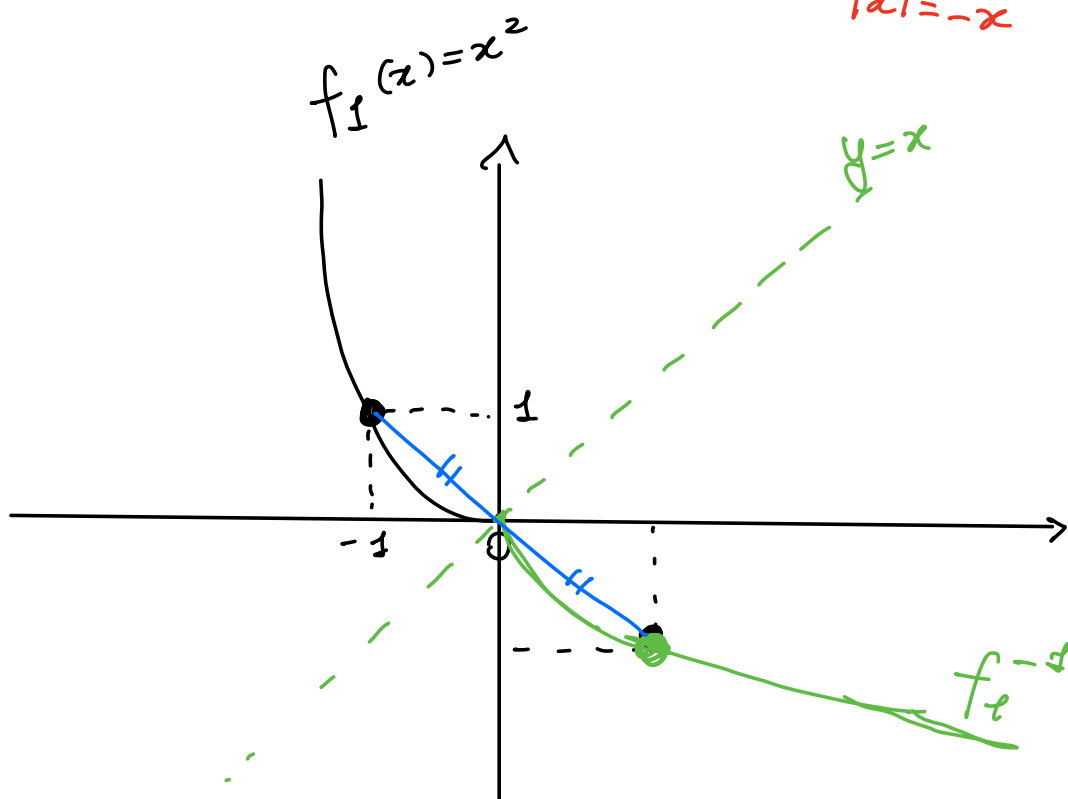
$$y = x^2$$

$$x = \pm \sqrt{y}$$

$$\text{N.B. } -\sqrt{y} = -\sqrt{x^2} \\ = -|x|$$

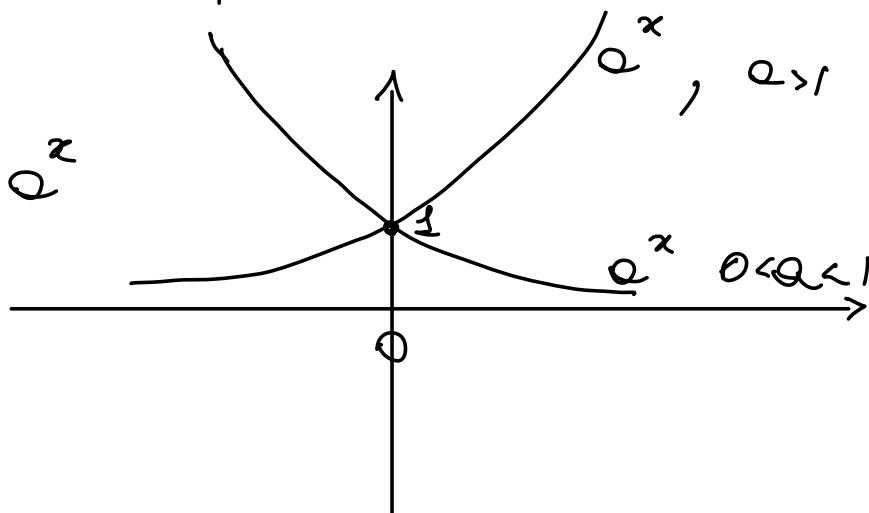
$$= -(-x) = x$$

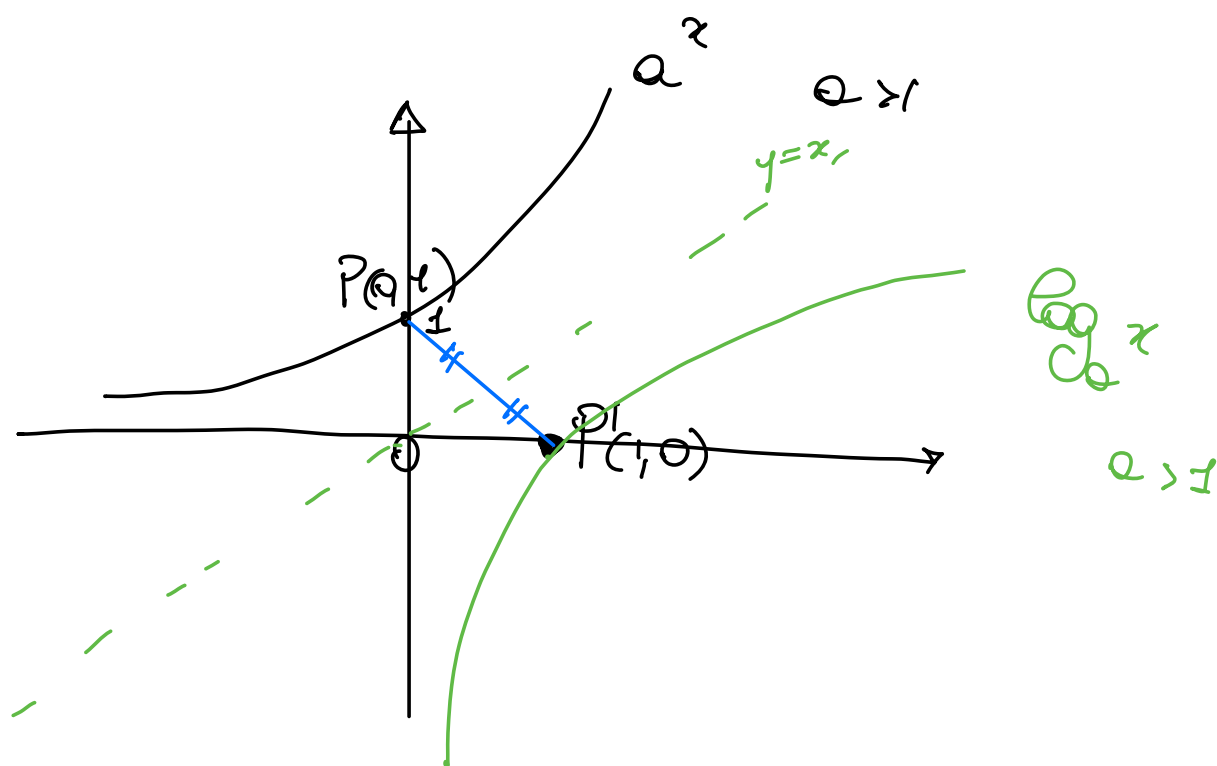
$$|x| = -x$$



ES

$$f(x) = a^x$$





x cosa trovare f_2^{-1}